

Project Title: Football Player Statistics Database System

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The purpose of this mini project was to design and develop a relational database system capable of storing, organizing, and analyzing football player statistics efficiently. The project was created using Microsoft Access for database management and Microsoft Excel for dashboard visualization.

The system was designed to improve the management of football player information by reducing redundancy, improving data accuracy, and allowing statistical analysis through queries, forms, reports, and dashboards.

The project demonstrates the use of relational databases, SQL queries, data validation, reporting systems, and data visualization techniques. It also highlights the importance of database systems within sports management environments where large volumes of player data must be managed effectively.

Tables

The screenshot above shows the relational tables used within the database system. Each table stores specific information relating to player statistics, positions, and performance records. Primary keys and foreign keys were used to maintain relationships and ensure data integrity.

The database was designed using multiple relational tables. Each table stored specific information to reduce duplication and improve organization.

Examples of tables used:

- Players Table
- Statistics Table
- Position Table
- Match Data Table

The use of separate tables improved normalization by ensuring data was stored efficiently and consistently.

Field Name	Data Type
Player_ID	Number
Player_Surname	Short Text
Player_Firstname	Short Text
Player_DOB	Date/Time
Player_Telephone	Short Text
Agent_ID	Number

General

Lookup

Field Size	Integer
Format	
Decimal Places	Auto
Input Mask	
Caption	
Default Value	0
Validation Rule	
Validation Text	
Required	Yes
Indexed	No
Text Align	General

Field Name	Data Type
Agent_ID	Number
Agent_Name	Short Text

General

Lookup

Field Size	Integer
Format	
Decimal Places	Auto
Input Mask	
Caption	
Default Value	0
Validation Rule	
Validation Text	
Required	Yes
Indexed	No
Text Align	General

Field Name	Data Type
Player_ID	Number
Position_ID	Number
Player_Position_Goal	Number
Player_Position_Yellow_Cards	Number
Player_Position_Rating	Number
Player_Position_Substitutions	Number

General

Lookup

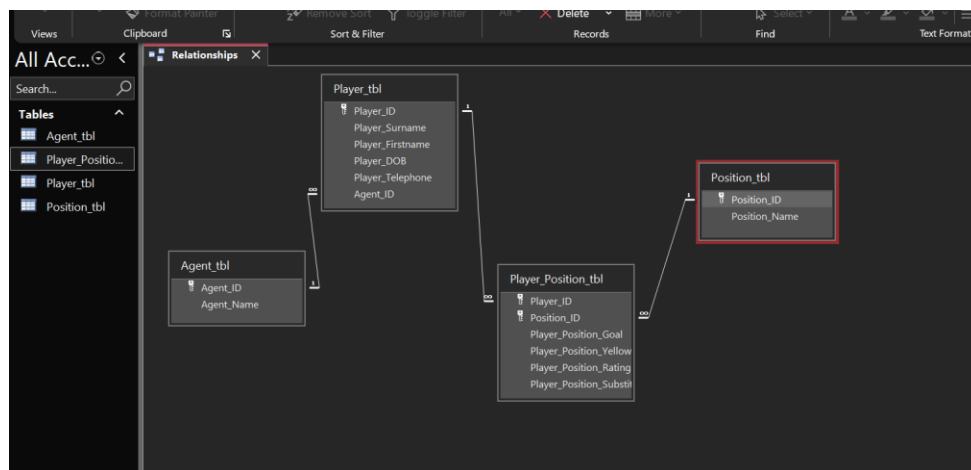
Field Size	Integer
Format	
Decimal Places	Auto
Input Mask	
Caption	
Default Value	0
Validation Rule	Between 1 And 5
Validation Text	
Required	No
Indexed	No
Text Align	General

	Field Name	Data Type	
	Position_ID	Number	v
	Position_Name	Short Text	

General Lookup	
Field Size	Integer
Format	
Decimal Places	Auto
Input Mask	
Caption	
Default Value	0
Validation Rule	
Validation Text	
Required	Yes
Indexed	Yes (No Duplicates)
Text Align	General

Relationship

The screenshot above demonstrates the relationships between the tables. One-to-many relationships were created to connect player records with their associated statistics and positions, reducing redundancy and improving consistency.



Query 1

Top Goal Scorers Query

Purpose:

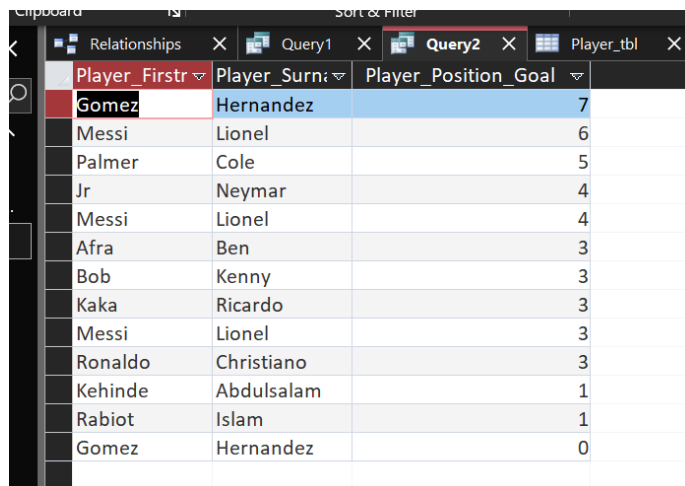
- To display players with the highest number of goals scored.

What it does:

- Displays player first name, surname, and goals scored.
- Sorts the goals in descending order, so the highest scorer appears first.

Why it is useful:

- Helps identify top-performing players quickly.
- Supports statistical analysis and performance monitoring.

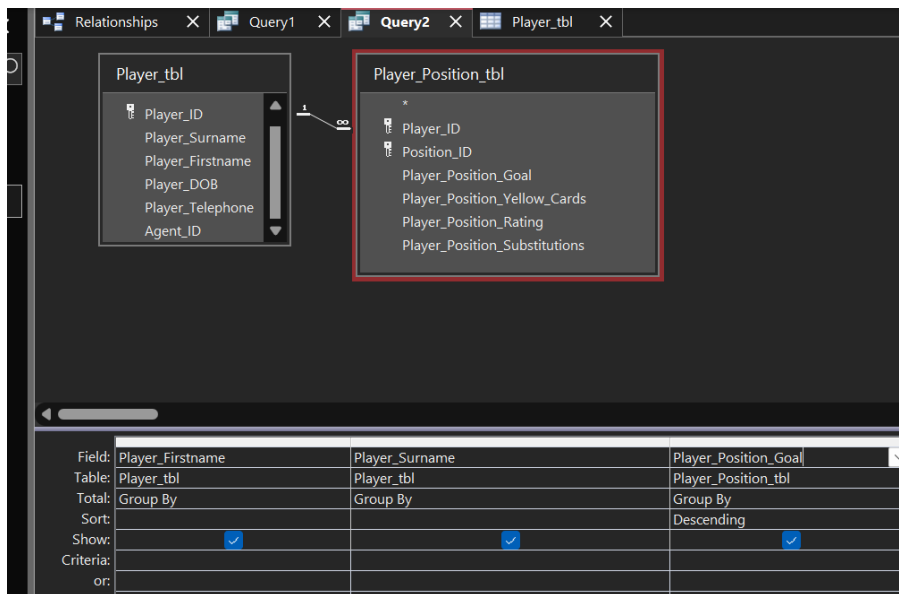


Player_FirstName	Player_Surname	Player_Position	Goals
Gomez	Hernandez		7
Messi	Lionel		6
Palmer	Cole		5
Jr	Neymar		4
Messi	Lionel		4
Afra	Ben		3
Bob	Kenny		3
Kaka	Ricardo		3
Messi	Lionel		3
Ronaldo	Christiano		3
Kehinde	Abdulsalam		1
Rabiot	Islam		1
Gomez	Hernandez		0

Design View

SQL

```
SELECT PlayerFirstName, PlayerSurname, Goals
FROM Players
ORDER BY Goals DESC;
```



Query 2: Age Statistics Query

Purpose:

- To analyze player statistics based on age groups.

What it does:

- Calculates player age.
- Counts the number of players within each age group.
- Calculates total yellow cards and goals for each age group.

Why it is useful:

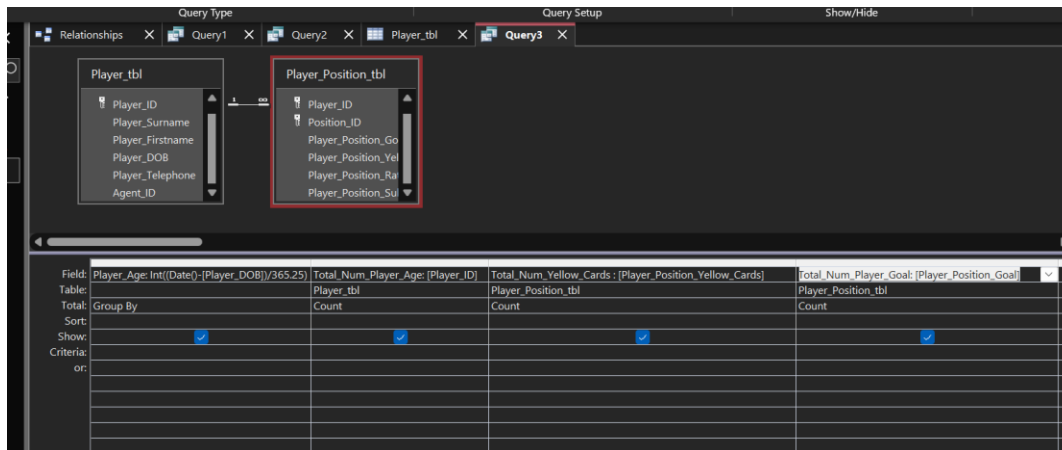
- Helps compare player performance by age.
- Supports recruitment and squad planning decisions.

Player_Age	Total_Num_Player_Age	Total_Num_Yellow_Cards	Total_Num_Player_Goal
17	1	1	1
18	3	3	3
19	1	1	1
20	3	3	3
21	2	2	2
22	3	3	3

Design View

SQL

```
SELECT Age,  
COUNT(PlayerID) AS TotalPlayers,  
SUM(YellowCards) AS TotalYellowCards,  
SUM(Goals) AS TotalGoals  
FROM Players  
GROUP BY Age;
```



Query 3

Create a flat file of the record

Player_ID	Player_Surname	Player_Firstname	Player_DOB	Player_Telephone	Position_ID	Position_Name	Player_Position_Goal	Player_Position_Yellow	Player_Position_Rating
1	Hernandez	Gomez	12/01/2006	(01234)1234	2	Midfielder	7	2	
2	Hernandez	Gomez	12/01/2006	(01234)1234	5	Right Back	0	0	
3	Islam	Rabiot	01/04/2007	(01234)2345	1	Goalkeeper	1	3	
4	Abdulsalam	Kehinde	10/07/2008	(01234)3456	2	Midfielder	1	1	
4	Lionel	Messi	06/12/2003	(01234)4567	3	Centre Forward	4	5	
4	Lionel	Messi	06/12/2003	(01234)4567	5	Right Back	6	7	
4	Lionel	Messi	06/12/2003	(01234)4567	6	Winger	3	4	
5	Ben	Afra	11/11/2005	(01234)5678	6	Winger	3	2	
6	Christiano	Ronaldo	25/02/2005	(01234)6789	2	Midfielder	3	4	
7	Neymar	Jr	30/09/2006	(01234)7890	4	Centre Back	4	3	
8	Cole	Palmer	22/07/2004	(01234)8901	4	Centre Back	5	3	
9	Kenny	Bob	14/04/2007	(01234)9012	3	Centre Forward	3	4	
10	Ricardo	Kaka	17/06/2007	(01234)0123	3	Centre Forward	3	4	

Player_Position_Goal	Player_Position_Yellow	Player_Position_Rating	Player_Position_Substitutions	Agent_ID	Agent_Name
7	2	3		1	Salam
0	0	1		8	1 Salam
1	3	4		1	2 James
1	1	5		0	1 Salam
4	5	2		3	3 Torres
6	7	3		2	3 Torres
3	4	5		3	3 Torres
3	2	4		5	2 James
3	4	5		2	3 Torres
4	3	5		2	2 James
5	3	4		2	1 Salam
3	4	3		0	2 James
3	4	5		3	1 Salam

Report

Multi-Position Player Query

Purpose:

- To identify players who can play more than one position.

What it does:

- Displays player name, surname, and date of birth.
- Calculates best rating across positions.
- Calculates total yellow cards in all positions.

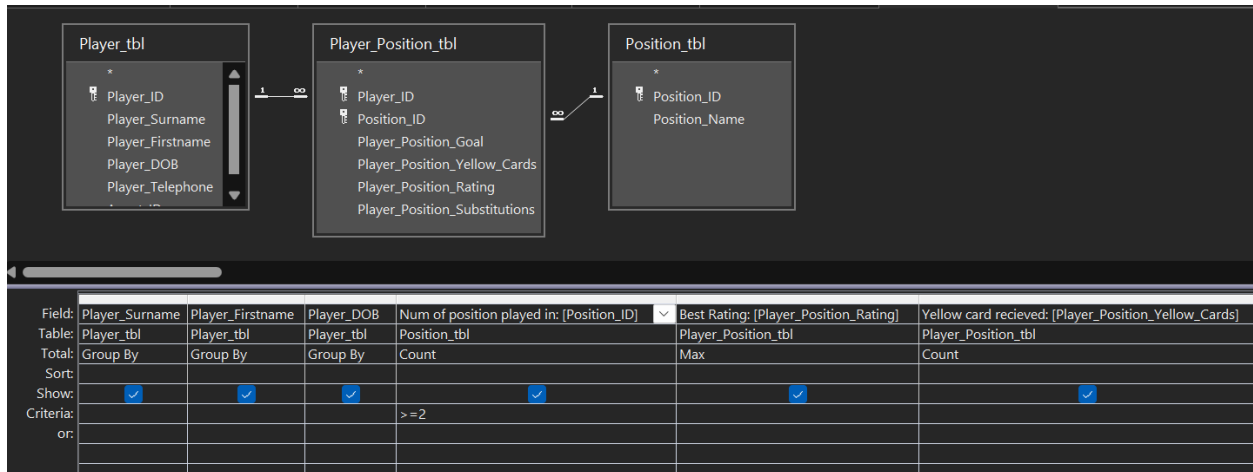
Why it is useful:

- Helps identify versatile players.
- Supports tactical and team selection decisions.

Query for report

SQL

```
SELECT PlayerSurname,  
PlayerFirstName,  
PlayerDOB,  
MAX(Rating) AS BestRating,  
SUM(YellowCards) AS TotalYellowCards  
FROM Players  
GROUP BY PlayerSurname, PlayerFirstName, PlayerDOB  
HAVING COUNT(Position) > 1;
```



Report Design view

Report Header																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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Report view

Multi-Position Players						
Multi-Position Players						
Player Surname	Player Firstname	Player DOB	Num of position played in	Best Rating	Yellow card recieved	
Hernandez	Gomez	12/01/2006	2	3	2	
Lionel	Messi	06/12/2003	3	5	3	
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Form

A ready to enter form to add new player

Form view

The screenshot above displays the data entry form used to add new player information. Validation rules and macros were implemented to ensure that required fields were completed accurately.

New Player Entry

Player ID	0	
Player Surname		(required)*
Player Firstname		(required)*
Player DOB		
Player Telephone		
Agent Name	▼ Salam James Torres	
Save Record		

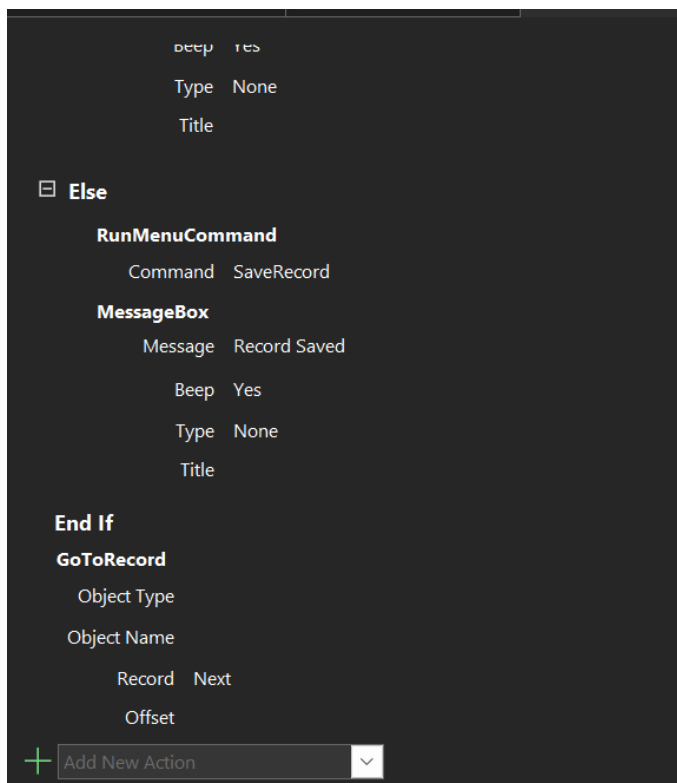
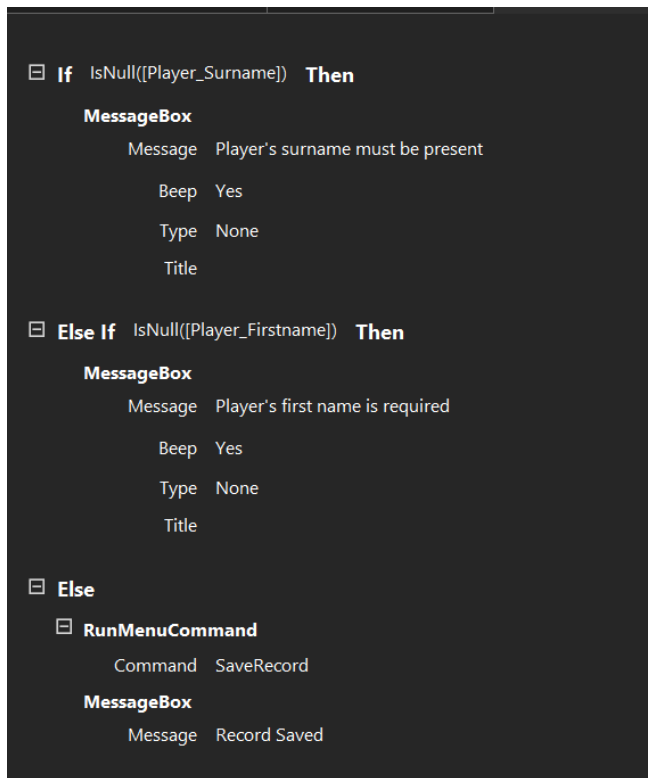
Form Marco and Validation

Macros were used to automate validation processes within the form.

The validation system ensured that:

- Required fields were completed
- Incorrect data formats were prevented
- Missing records could not be submitted

This improved data reliability and reduced the possibility of human error.



Design view

1234567891011

Form Header

New Player Entry

Detail

Player ID

Player ID

Player Surname

Player Surname

Player Firstname

Player Firstname

Player DOB

Player DOB

Player Telephone

Player Teleph

Agent Name

Unbound

Save Record

Form Footer

1234567891011121314

Form Header

New Player Entry

Detail

Player ID

Player ID

Player Surname

Player Surname

Player Firstname

Player Firstname

Player DOB

Player DOB

Player Telephone

Player Teleph

Agent Name

Unbound

Save Record

Form Footer

Property sheet

Selection type: Text Box

Player_Telephone

Format

Data

Event

Other

All

Control Source

Player_Telephone

Text Format

Plain Text

Input Mask

\(0"1234)"0000;_

Default Value

Validation Rule

Validation Text

Filter Lookup

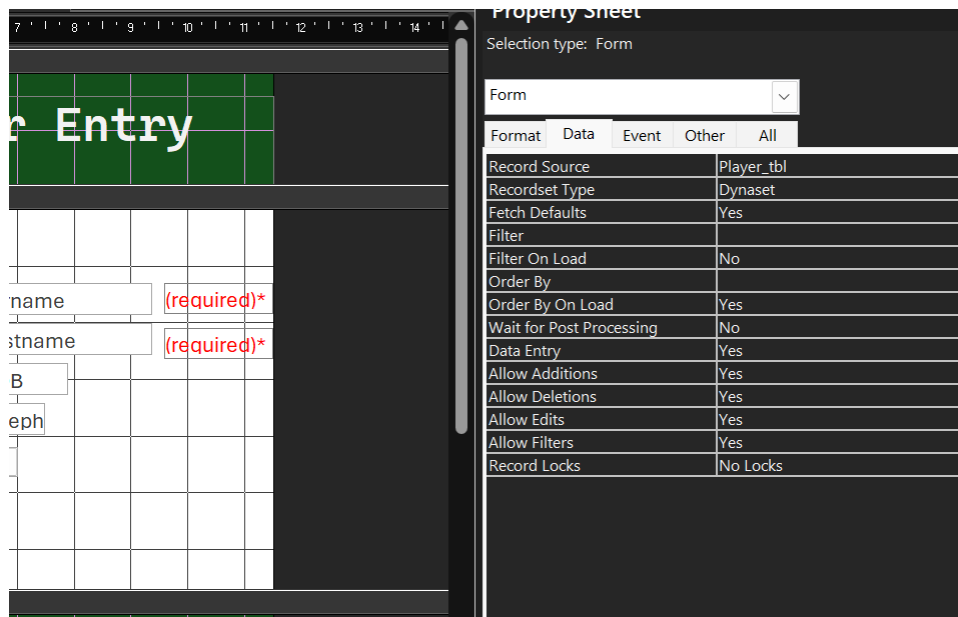
Database Default

Enabled

Yes

Locked

No



Form Entry

The image shows a 'New Player Entry' form in Microsoft Access. The form has a green header with the title 'New Player Entry'. Below the header, there are several input fields:

- Player ID: 99
- Player Surname: Abdulsalam (required)*
- Player Firstname: Kehinde (required)*
- Player DOB: 10/07/2008
- Player Telephone: (01234)7777
- Agent Name: Salam (dropdown menu)

At the bottom of the form is a blue 'Save Record' button. A 'Microsoft Access' dialog box is open, displaying 'Record Saved' and an 'OK' button.

Excel

I used query to pick all table titles to create a flat file and export the data from access to Microsoft excel to create dashboard for the player statistic.

The database data was exported into Microsoft Excel to create an interactive dashboard.

Features included:

- Pivot tables
- KPI indicators
- Charts
- GETPIVOTDATA functions
- Data visualization tools

Benefit of Dashboard

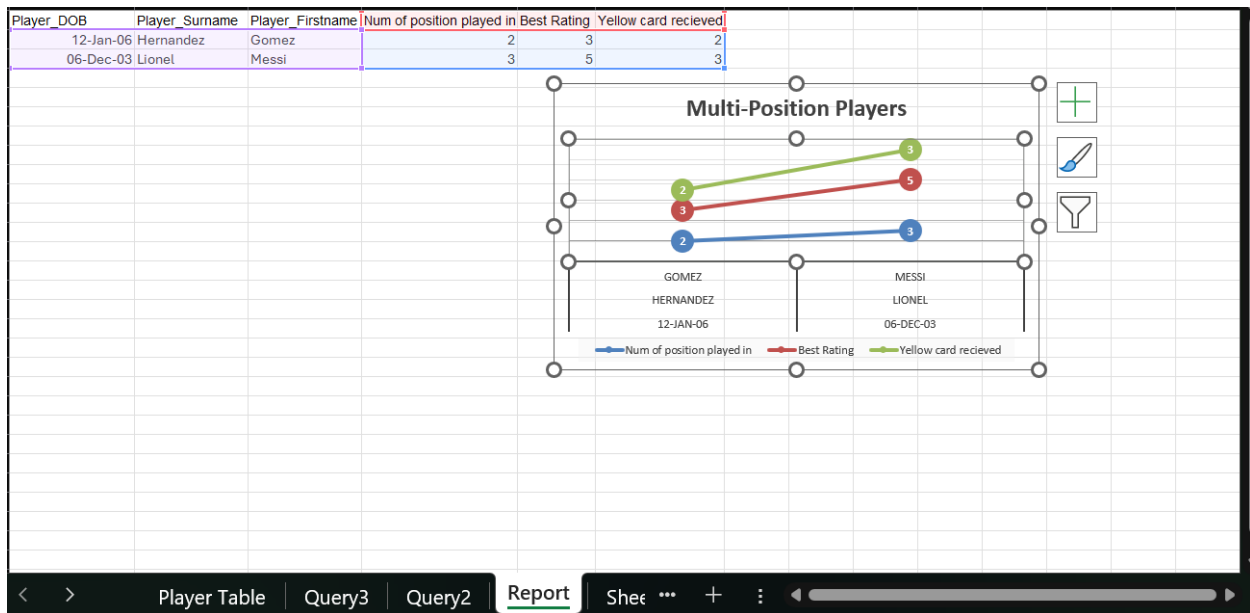
- Improved data interpretation
- Easier performance analysis
- Visual representation of statistics
- Professional presentation of data

Dashboard

The screenshot above shows the Excel dashboard created from the exported database data. Pivot tables, KPI indicators, and charts were used to present player statistics visually and improve data interpretation



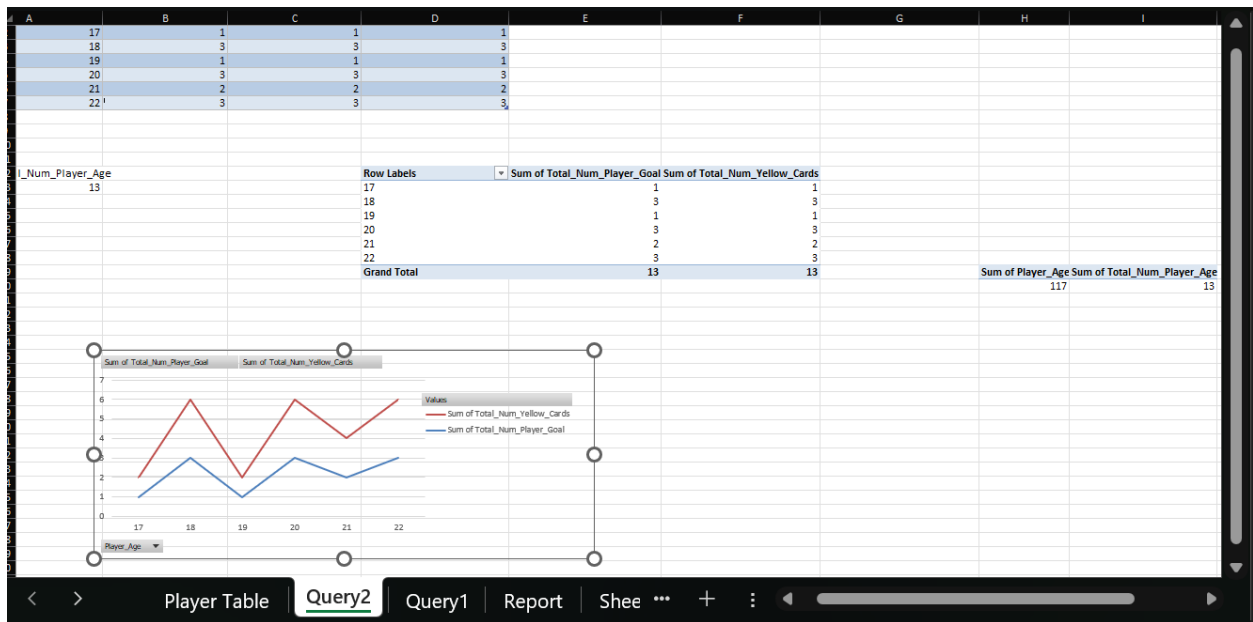
Report sheet



Query1: Sheet For Goal Scoring Descending Order

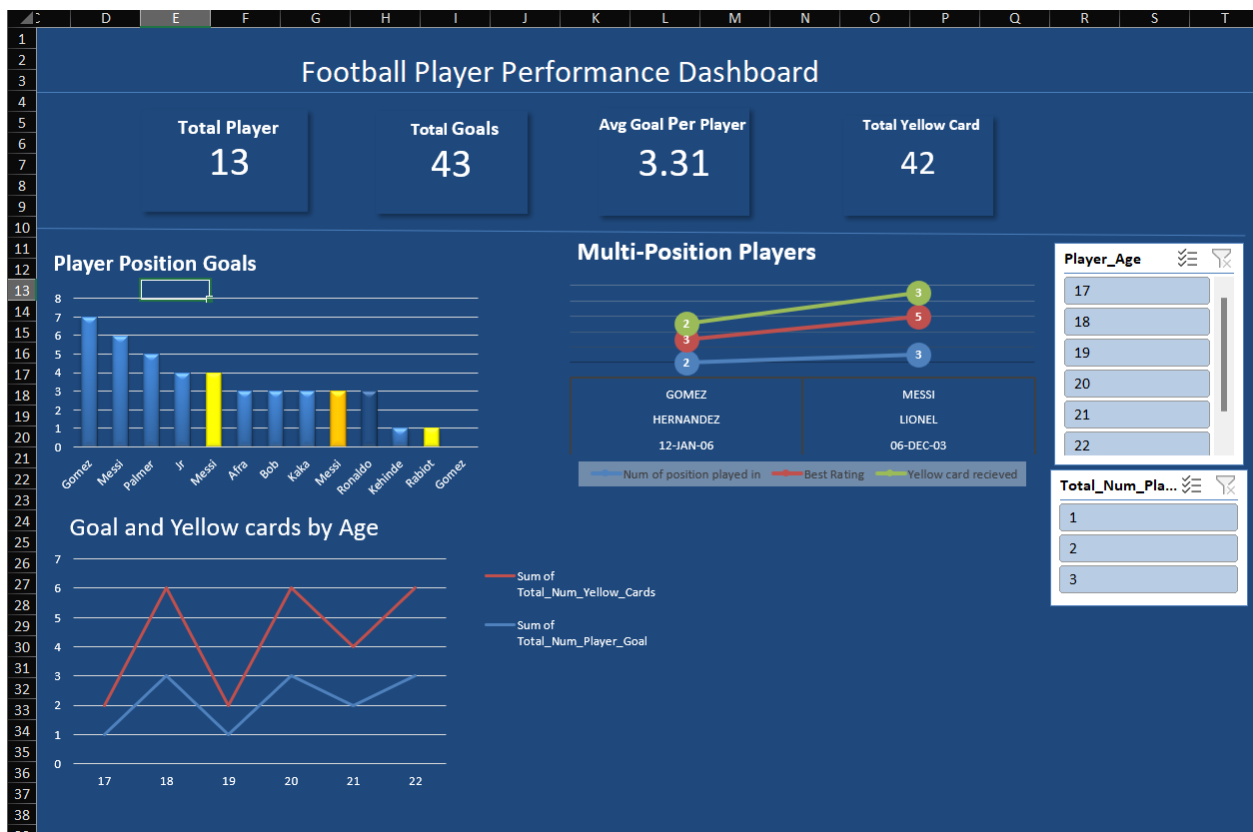


Query 2



Excel Dashboard Optimization

Creating a KPI button for the for enhance using the shape and text and getpivotdata function to bring the value to the dashboard and making it more eye pleasing



Evaluation

The project was successful in creating an organized and efficient database management system for football player statistics. The use of relational tables improved data organization and reduced redundancy throughout the system.

The relationships between the tables enhanced data integrity by ensuring that player statistics were linked correctly to player records. This improved consistency and reduced duplication of information.

The queries successfully analyzed the stored data and generated meaningful information. The goal scorer query effectively sorted players according to performance, while the age analysis query provided useful statistical comparisons between age groups.

Forms significantly improved usability because users could enter data through a user-friendly interface instead of directly interacting with tables. Validation rules and macros improved accuracy by reducing the likelihood of incomplete or incorrect data entry.

Reports enhanced the professional presentation of information and made it easier to interpret player statistics. The Excel dashboard further improved the project by transforming raw data into visual charts and KPI indicators that supported decision-making.

One weakness of the system is that it relies heavily on manual data entry, which increases the possibility of human error. Additionally, the database currently lacks advanced security features such as user authentication and automated backup systems.

Despite these limitations, the project achieved its objectives successfully and demonstrated a strong understanding of relational database systems, SQL queries, data analysis, and dashboard development.

Strength of Project

- Strong relational database structure
- Effective use of SQL queries
- User-friendly forms
- Professional reporting system
- Interactive Excel dashboard
- Reduced data redundancy
- Improved data integrity
- Good use of validation and macros

Weakness of the Project

- Manual data entry process
- Limited automation
- No advanced security system
- No cloud integration
- Limited real-time analytics

Future Improvement and Optimisation

Several improvements could be implemented in the future to enhance the performance and functionality of the database system.

One improvement would be implementing user authentication and access control. This would improve security by ensuring only authorised users can